

ANATOMY

Exploring the Human Body



Dr. Pradeep Pawar

Preface

Vision of the Notes

“The Study of Anatomy is the Beginning of the Understanding of Man.”
— Andreas Vesalius (Father of Modern Anatomy)

Welcome to the Enchanting World of Anatomy!

Anatomy serves as the bedrock of medical education, providing a thorough understanding of the structure and organization of the human body—an essential foundation for accurate diagnosis and appropriate patient care.

These notes have been meticulously crafted to support your postgraduate preparation journey for exams such as NEET-PG, INI-CET, NExT, and FMGE. Designed to complement your lectures and standard textbooks, they offer clear, concise explanations, detailed illustrations, and clinical correlations to enrich your learning experience.

A special focus has been placed on clinically significant concepts, enabling you to integrate theoretical knowledge with practical applications. Whether you're revising or solving MCQs, these notes are tailored to help you think critically and apply your understanding effectively.

We've included a wide array of labeled diagrams, high-quality visuals, and 3D models to support visual learning. These tools will aid in navigating the complex details of human anatomy and foster a deeper grasp of spatial relationships and structural intricacies.

These notes are more than just a study aid—they are a reliable companion throughout your medical journey. We encourage you to embrace challenges, celebrate progress, and always remember the profound impact your anatomical knowledge will have on your future clinical practice and on your patients.

The human body is an extraordinary system of interlinked structures. As you explore its wonders, may this foundation empower you to become a skilled, compassionate, and confident healthcare professional. Keep exploring!

“Anatomy is not just a subject to memorize—it's a map to navigate life-saving decisions.”

Wishing You Success and Inspiration on Your Anatomical Adventure!

Features of the Notes

These anatomy notes have been designed with precision and purpose to support your PG exam preparation.

Key features include:

- 1. Integrated with MedEd Lectures**
Aligned to complement the lecture videos available on the MedEd application for seamless learning.
- 2. Comprehensive Coverage**
Covers all essential topics with depth and breadth suitable for NEET-PG, INI-CET, NExT, and FMGE aspirants.
- 3. Structured and Updated Content**
Organized in a logical, easy-to-follow manner, based on the latest evidence and exam trends.
- 4. Clear, Conceptual Explanations**
Focuses on clarity and concept-building, simplifying complex topics for better retention.
- 5. Detailed Illustrations and Labeled Diagrams**
Rich visuals aid in understanding anatomical structures and their spatial relationships.
- 6. Clinical Relevance**
Highlights key clinical correlations to bridge the gap between basic anatomy and medical practice.
- 7. Key Points Highlighted**
Important facts and takeaways are emphasized to make revision more efficient.

How to Use the Notes Efficiently:

To maximize the benefits of your printed anatomy notes, follow these strategies:

- 1. Highlight and Underline Key Concepts**
Use color-coded highlighters or underlining to draw attention to important terms, structures, definitions, and clinical correlations for easy revision.
- 2. Add Handwritten Annotations**
While watching lecture videos on the MedEd app, jot down additional notes, mnemonics, or memory aids in the margins. Handwritten annotations boost memory and deepen understanding.
- 3. Engage in Active Reading**
Don't just read—interact with the content. Ask yourself questions, summarize sections in your own words, and try to recall key points without looking.

4. Review Regularly and Consistently

Set aside time each week to revisit your notes and re-watch videos. Regular revision strengthens retention and builds long-term memory.

5. Practice with Question Banks

Use these notes as a reference while solving MCQs or mock exams on the PW MedEd app. Applying what you've learned helps reinforce concepts and identify weak areas.

About the Faculty

Dr. Pradeep Pawar, MBBS, MS (Human Anatomy) with nearly two decades of teaching experience, has guided thousands of undergraduate students across medical, dental, physiotherapy, and nursing streams, as well as mentored aspirants preparing for postgraduate entrance exams like NEET-PG, INI-CET, FMGE, and USMLE.

He is also a recognized guide for MS and PhD programs in Anatomy. Renowned for his conceptual teaching style, Dr. Pawar is passionate about simplifying the subject, building strong foundational knowledge, and making anatomy enjoyable. He is widely appreciated by students and maintains an active presence on social media platforms, where he continues to inspire and educate.

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1. Dr. F. Anto Nazarene

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About PW MedEd

In the rapidly evolving field of medicine, lifelong learning and continuous professional development are critical for delivering quality healthcare. With this vision, PW (Physics Wallah)—India's leading ed-tech platform—launched PW MedEd, a dedicated initiative for NEET PG, INI-CET, NExT, and FMGE preparation.

PW MedEd is designed to be a one-stop solution for medical aspirants, offering comprehensive, high-quality content at affordable prices.

The platform features:

- ☞ A team of 19 expert faculty members
- ☞ Over 300 practicing doctors from leading institutions
- ☞ High-quality video lectures
- ☞ Video-based notes
- ☞ A dynamic Q-Bank
- ☞ Full-length test series & subjectwise tests
- ☞ Live and interactive classes
- ☞ 3D Models & Slides
- ☞ For Assistance & Clarification of doubts & queries : AI Guru (24/7)
- ☞ Many more innovative ingredients

PW MedEd aims to create a thriving, supportive community of future specialists, equipping them with the tools they need to succeed. With a strong emphasis on affordability, accessibility, and academic excellence, PW MedEd is playing a pivotal role in transforming medical education in India, and empowering tomorrow's healthcare leaders.

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Introduction to Anatomy

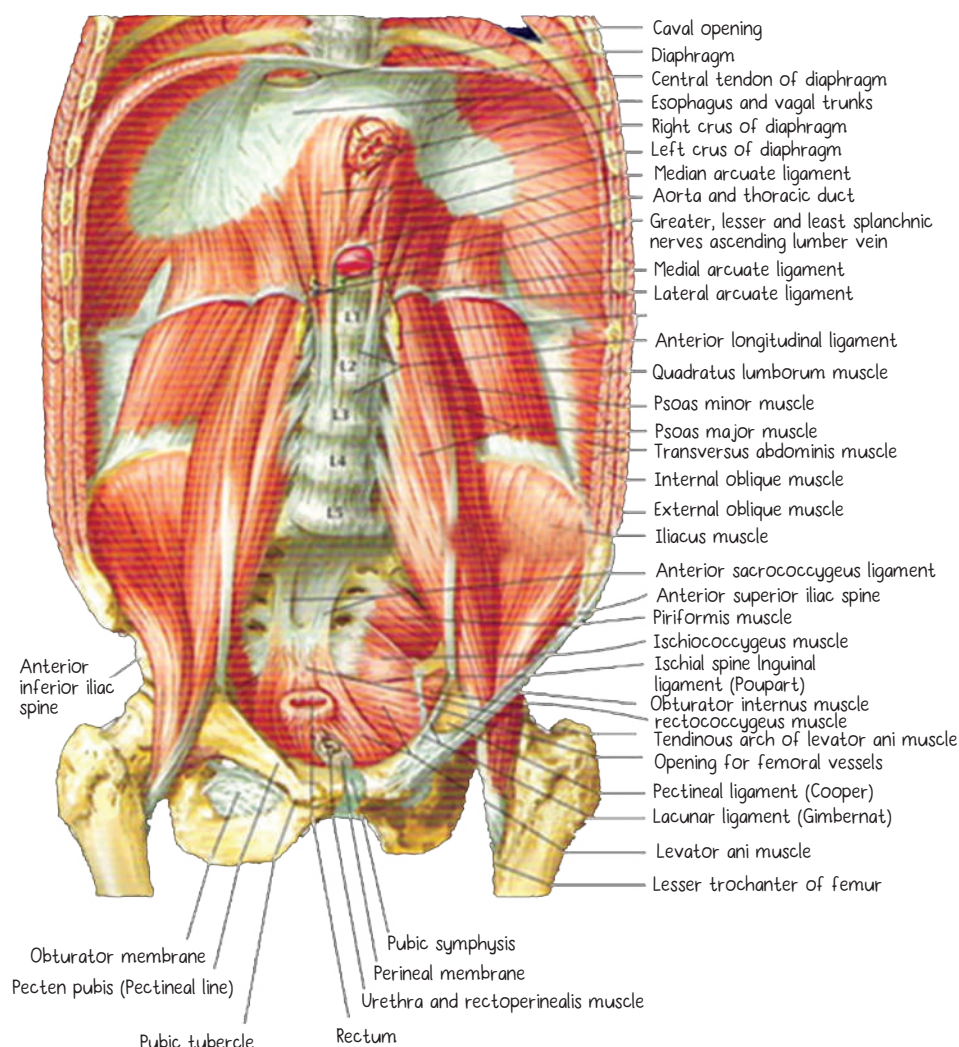
- Human anatomy is the science which deals with **structure** of the human body.
- The term anatomy is derived from Greek word "**Anatome**" means "cutting up".
- Anatomy is to physiology what geography is to history.
- It describes the theater in which the action takes place.

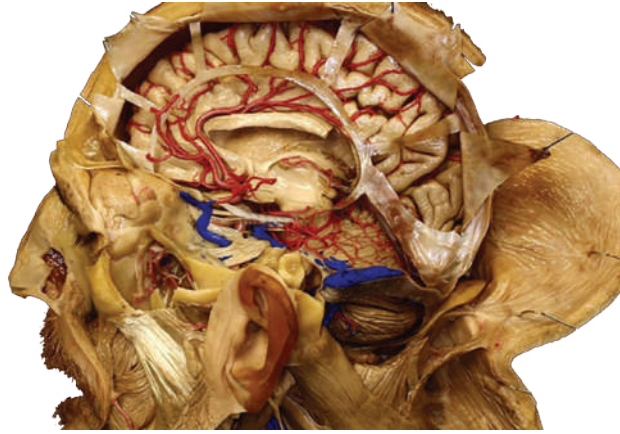
SUBDIVISIONS OF ANATOMY

Cadaveric Anatomy/Macroscopic Anatomy

- Studied on dead bodies with the naked eye
- Gross/macroscopic anatomy
- This includes:
 - a. **Regional** anatomy
 - b. **Systemic** anatomy

Personal Notes





Cadaveric Head - Dissection

Living Anatomy

- It is studied on living beings.
- It is studied by palpation, percussion and auscultation.



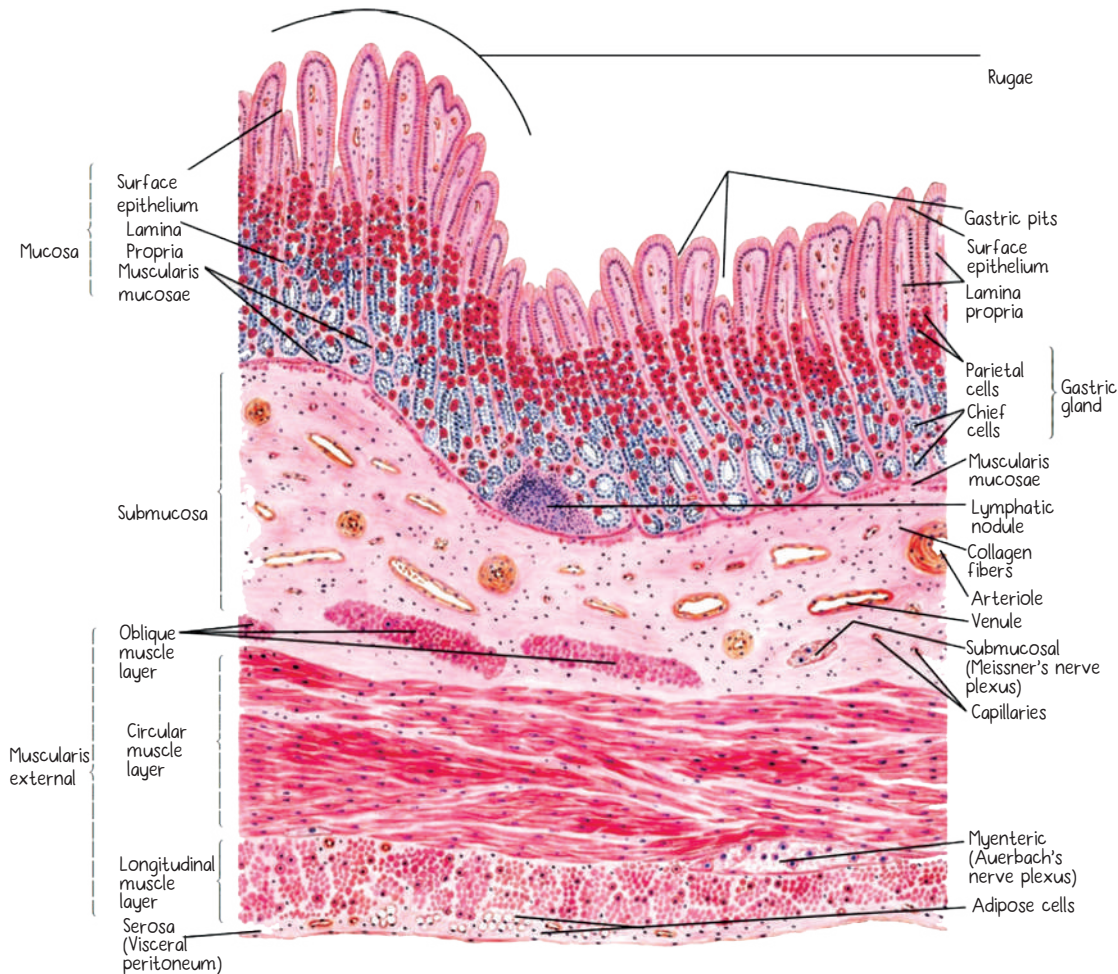
Developmental Anatomy/Embryology

- It is the study of prenatal and postnatal changes in an individual.



Microscopic Anatomy

- It is the study of structures with the aid of a microscope.



Personal Notes

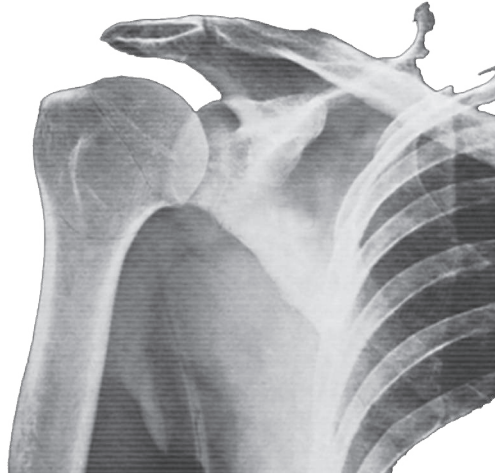
Surface/Topographic Anatomy

- It is the study of the deeper part of the body in relation to the skin surface.



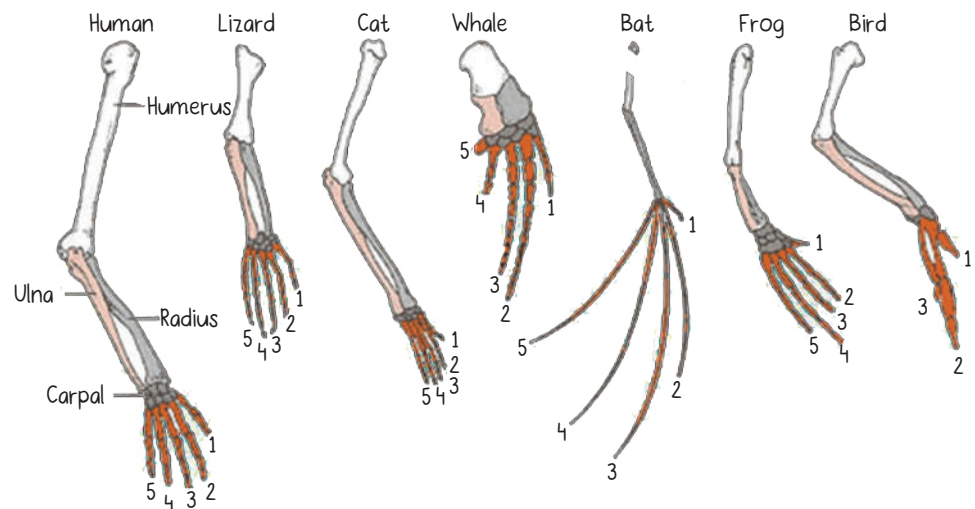
Radiographic Anatomy

- It is the study of deeper structures by plain or contrast radiography.



Comparative Anatomy

- It is the study of anatomy of other animals.
- It explains changes in form, structure and function of different parts of the human body.



Applied Anatomy

- It deals with the application of anatomical knowledge to medical and surgical practice.

POSITION OF BODY

Anatomical Position

1. Body is erect
2. Eyes look straight in front
3. Upper limbs are hanging by the side of the trunk with palms facing forward.

4. Lower limbs are parallel with toes pointing forward.



Supine Position

- Lying down position with face directed upwards.

Prone Position

- Lying down position with face directed downwards.



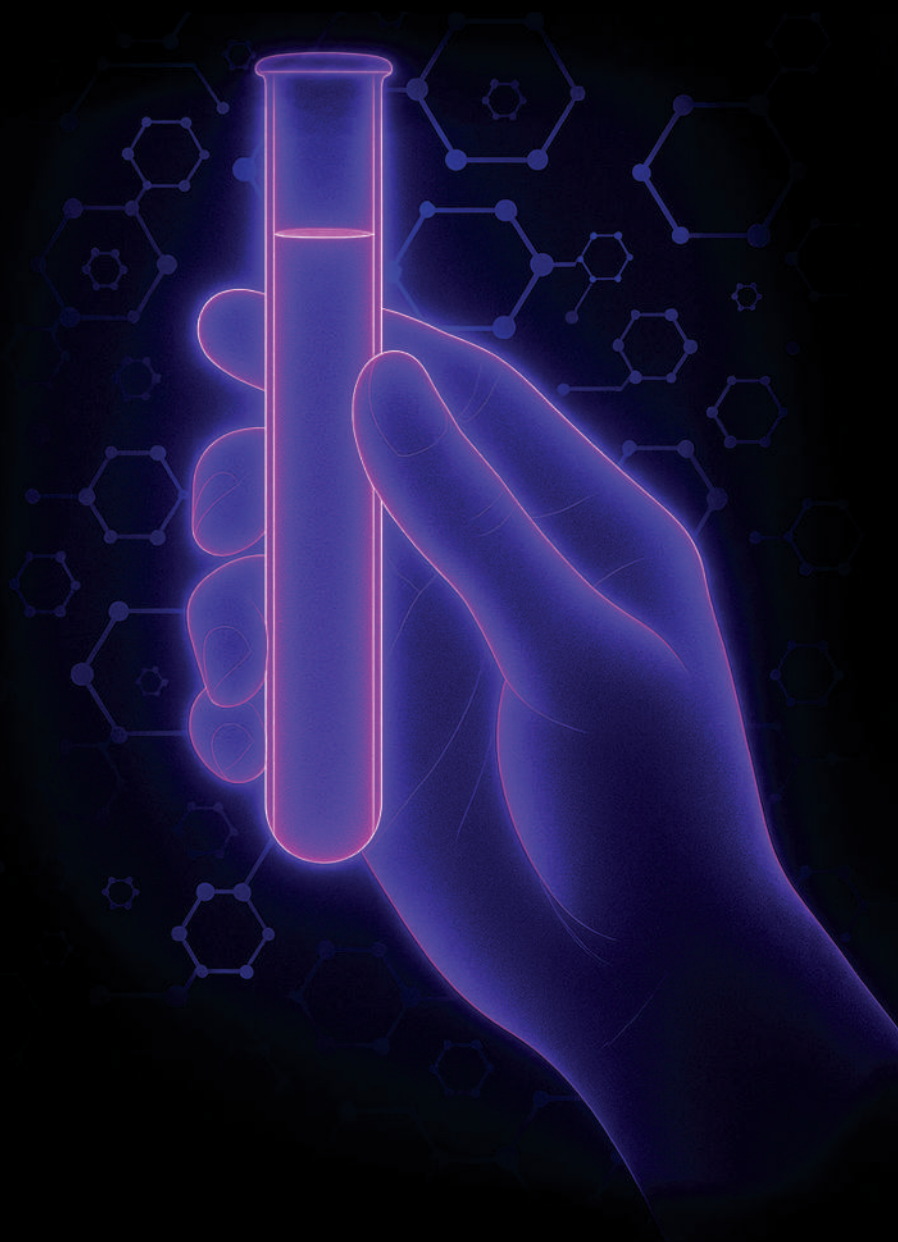
Lithotomy Position

1. Lying supine with buttocks at the edge of the table.
2. Hips and knees fully flexed.
3. Feet strapped in position.



BIOCHEMISTRY

Molecules of Life



Dr. Rajesh Jambhulkar

About the Faculty

Dr. Rajesh K. Jambhulkar, MBBS, MD, DNB (Biochemistry), is a distinguished Medical Educator and Associate Professor of Biochemistry at a reputed Government Medical College in Maharashtra, India. With over 13 years of dedicated teaching experience, Dr. Jambhulkar has made significant contributions to undergraduate medical and paramedical education. A passionate teacher and mentor, he is widely recognized for his ability to simplify complex biochemical concepts, making them accessible and engaging for students.

Beyond the walls of the classroom, Dr. Jambhulkar has extended his educational reach through digital platforms – most notably YouTube – where he serves as a trusted academic guide to a growing community of over 5.5 lakh learners across India and beyond. Renowned for his clarity, commitment, and student-centric approach, Dr. Jambhulkar continues to inspire the new generation of medical professionals.

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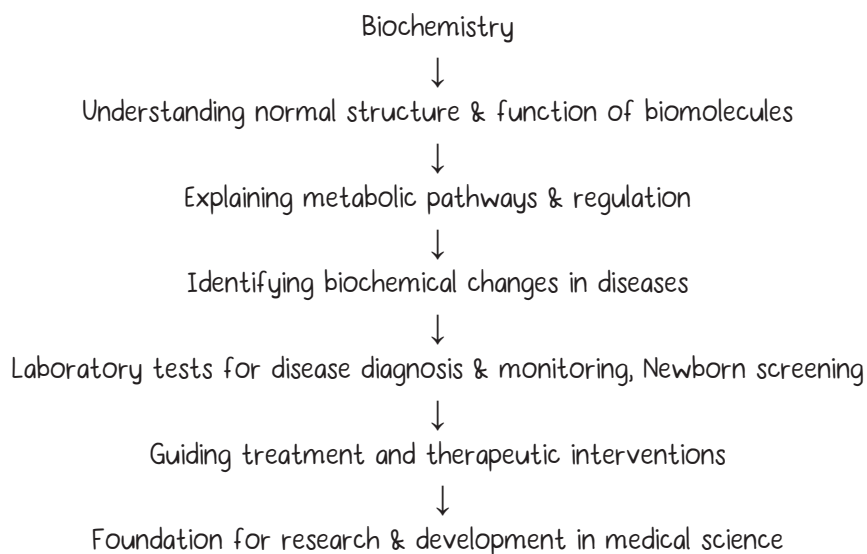
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Introduction of Biochemistry & Important Topics

INTRODUCTION

Biochemistry is the branch of science that explores the chemical processes within and related to living organisms. It is a laboratory-based science combining biology and chemistry, focusing on molecular mechanisms that control cellular function.

Biochemistry = **Language of Medicine** → It explains the molecular basis of life, detects disease changes, guides diagnosis & therapy, and integrates basic science with clinical practice.



 Personal Notes

CARBOHYDRATES

- **Key Molecules:** Glucose, Fructose, Galactose
- **Associated Disorder:** Diabetes Mellitus
 - ▶ **Features:** Polyuria, Polydipsia, Polyphagia
 - ▶ **Laboratory Diagnosis:**
 - Fasting/Random/Postprandial Blood Glucose
 - HbA1C (8 - 12 weeks average)
 - Fructosamine (2 - 3 weeks average)

LIPIDS

- **Key Molecules:** Triglycerides, Cholesterol, Lipoproteins, Phospholipids, Sphingolipids
- **Disorders:**
 - ▶ Atherosclerosis
 - ▶ Fatty Liver

- ▶ Coronary Artery Disease
- ▶ Respiratory Distress Syndrome
- **Tests:** Lipid Profile: TG, Cholesterol, HDL, LDL, VLDL

PROTEINS AND HEMOGLOBIN

- Amino acids classification
- Structural organization
- **Diseases:**
 - ▶ Hemoglobinopathies: Sickle Cell Disease, Thalassemia
 - ▶ Inborn Errors: PKU, Alkaptonuria, Homocystinuria
- **Diagnosis:** Electrophoresis (For SCD, thalassemia)

NUCLEIC ACIDS

- **Role:** Genetic material (DNA, RNA)
- **Associated Disorders:** Mutations, Genetic Disorders
- **Diagnostic Tools:** PCR, DNA Chips, Blotting Techniques

IMPORTANT TOPICS OVERVIEW

- Vitamins (Fat-soluble, Water-soluble)
- Minerals (Zn, Cu, Fe, Iodine, etc.)
- Metabolic Disorders (PKU, MSUD, GSD)
- Enzymes (Kinetics, Inhibitors, Diagnostic markers)
- Acid-Base Balance
- Genetics (DNA replication, mutation, gene expression)
- Biological **Oxidation/ETC/OP**

VITAMINS OVERVIEW

Fat-Soluble Vitamins (A, D, E, K)

Vitamin	Function	Deficiency Manifestations
A	Vision (Rhodopsin synthesis), epithelial integrity	Night blindness, Xerophthalmia, Conjunctival xerosis, Bitot spots, Corneal ulceration, Phrynoderma
D	Calcium-phosphorus homeostasis, bone mineralization	Rickets (Children), Osteomalacia (Adults); ↑ ALP
E	Antioxidant; membrane protection	Rare; Hemolytic anemia in neonates, neurological symptoms in adults
K	Post-translational γ -carboxylation of clotting factors II, VII, IX, X	Bleeding disorders
	Forms: K1 (Phylloquinone), K2 (Menaquinone), K3 (Menadione)	K3 is synthetic and water-soluble; can cause hemolysis in neonates if overdosed

Water-Soluble Vitamins (B-Complex, C)

Personal Notes

Vitamin	Function	Deficiency Manifestations
B1 (Thiamine)	Oxidative decarboxylation (e.g. Pyruvate DH, α -KG DH)	Beriberi (Dry, wet), Wernicke-Korsakoff syndrome
B2 (Riboflavin)	Component of FAD (ETC, redox reactions)	Ariboflavinosis: Cheilitis, glossitis, seborrheic dermatitis
B3 (Niacin)	NAD ⁺ /NADP ⁺ coenzyme for redox reactions	Pellagra (3 D's: Dermatitis, Diarrhea, Dementia)
B5 (Pantothenic acid)	Acyl carrier (CoA synthesis)	Burning feet syndrome (Rare)
B6 (Pyridoxine)	Transamination, decarboxylation, neurotransmitter synthesis	Neuropathy, irritability, convulsions (Especially with isoniazid use)
B7 (Biotin)	Carboxylation reactions (e.g. Pyruvate carboxylase, acetyl-CoA carboxylase)	Dermatitis, alopecia, enteritis; seen in raw egg white (Avidin) excess
B9 (Folic acid)	One-carbon transfers, nucleotide synthesis	Macrocytic anemia, NTD in fetus, megaloblastic anemia
B12 (Cobalamin)	Methylmalonyl-CoA mutase, Methionine synthase	Macrocytic anemia, Subacute Combined Degeneration of the spinal cord
C (Ascorbic acid)	Hydroxylation of proline/lysine in collagen, antioxidant	Scurvy: Bleeding gums, poor wound healing, perifollicular hemorrhages

MINERALS AND THEIR DISORDERS

Mineral	Associated Disorder(s)	Key Enzyme(s)/Function
Calcium	Tetany (Trousseau's & Chvostek's signs)	Muscle contraction, blood coagulation, signal transduction
Zinc	Acrodermatitis enteropathica	Cofactor for carbonic anhydrase, Superoxide Dismutase (SOD)
Copper	Wilson's disease, Menke's disease	Cofactor for tyrosinase, lysyl oxidase, cytochrome c oxidase
Iodine	Goitre, Cretinism	Required for thyroid hormone synthesis (T3, T4)
Fluoride	Dental and skeletal fluorosis	Strengthens enamel; inhibits enolase in glycolysis
Iron	Iron Deficiency Anemia (IDA)	Essential for cytochromes, hemoglobin, catalase
Selenium	Keshan disease (Cardiomyopathy)	Cofactor for glutathione peroxidase, maintains redox balance

CARBOHYDRATE METABOLISM [BC 3.1 TO BC 3.6]

GAGs & MPS (Mucopolysaccharidoses)

- Hurler's Syndrome: α -L-iduronidase deficiency
- Hunter's Syndrome: Iduronate-2-sulfatase deficiency.

- Heparan Sulfate:
 - ▶ Anchors Lipoprotein Lipase (LPL)
 - ▶ Contributes to glomerular capillary charge and selective permeability

Glycogen Storage Diseases

- Type I (Von Gierke): ↓ Glucose-6-phosphatase → severe fasting hypoglycemia
- Type II (Pompe): ↓ Lysosomal acid maltase → cardiomegaly, hypotonia
- Type III (Cori): ↓ Debranching enzyme → mild hypoglycemia, muscle weakness
- Type IV (Andersen): ↓ Branching enzyme → cirrhosis, early death
- Type V (McArdle): ↓ Muscle phosphorylase → exercise cramps, myoglobinuria
- Type VI (Hers): ↓ Liver phosphorylase → mild hypoglycemia, hepatomegaly

Other Disorders

- Galactosemia: GALT deficiency → dulcitol buildup, “oil-drop cataract”
- Lactose Intolerance: Lactase deficiency
- DKA (Diabetic Ketoacidosis): Ketone body excess, metabolic acidosis
- Pentosuria: Benign presence of xylulose in urine

Key Metabolic Pathways

- Glycolysis: 3 irreversible steps; Fluoride inhibits enolase
- Gluconeogenesis: 4 irreversible enzymes, 4 key substrates
- HMP Shunt: Generates NADPH, ribose sugars
- Rapoport-Luebering Cycle: Produces 2,3-BPG for oxygen unloading
- Pyruvate Dehydrogenase Complex: 3 enzymes, 5 coenzymes (TPP, lipoic acid, CoA, FAD, NAD⁺)

LIPID METABOLISM [BC 4.1 TO BC 4.8]

Disorders

- Refsum's Disease: Phytanic acid accumulation (peroxisomal disorder)
- SIDS: Carnitine/acyl-CoA dehydrogenase deficiency
- Jamaican Vomiting Sickness: Hypoglycin A toxicity
- Fatty Liver Disease
- Respiratory Distress Syndrome: Surfactant deficiency (↓ Dipalmitoyl lecithin)

Key Topics

- Beta-Oxidation
- Fatty Acid Synthase (FAS) Complex
- Ketone Body Metabolism
- Lipoprotein Metabolism (Chylomicrons, VLDL, LDL, HDL)

 MedEd⁺ Edition 3.0



PHYSIOLOGY

The Marvels of Life



Dr. Vivek Nalgirkar

About the Faculty

Dr. Vivek Nalgirkar, MBBS, MD (Physiology) is an esteemed academician who holds an MBBS and an MD in Physiology. With over two decades of teaching experience, he has been a dedicated educator for undergraduate students across various disciplines—including medicine, dentistry, physiotherapy, and nursing.

In addition to his work with undergraduates, Dr. Nalgirkar has played a pivotal role in guiding postgraduate medical aspirants, offering expert coaching for PG entrance examinations. He also serves as a mentor and research guide for MD and PhD students, shaping the next generation of medical professionals and researchers. A committed scholar, he has authored three well-regarded books on physiology, further contributing to the academic literature and supporting learners with comprehensive, accessible resources.

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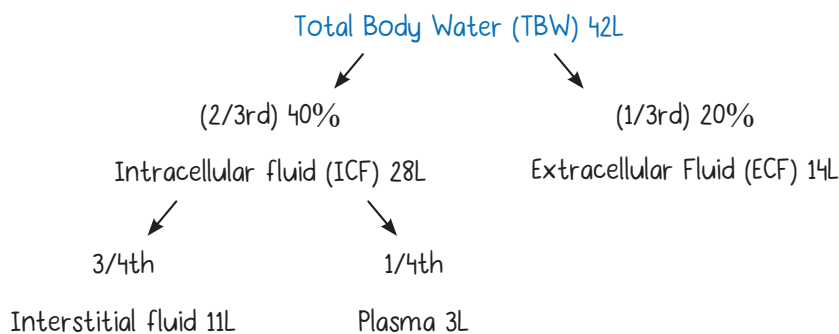
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Body Fluid Compartments

VOLUMES

- Total Body Water (TBW) = 60% of the body weight (Reference man is 70 kg, healthy adult male)



- Cellular Component (3% of Body weight)
- Blood (8% of Body weight = 5L)

Newborns are More Prone to Dehydration

- Greater metabolic rate
- Inability to express thirst.
- Higher evaporative loss due to a greater surface area-to-volume ratio.
- Approximately 1L of extracellular fluid belongs to the **transcellular fluid compartment** which is enclosed as CSF, synovial fluid, pleural fluid, pericardial fluid, peritoneal fluid and fluid in various spaces.

MEASUREMENTS OF BODY FLUID COMPARTMENTS

Conventional Method

- Indicator Dilution Method/Stewart: Hamilton Method/Dye Dilution Method

Steps

- Selection of indicator

Ideal Property

1. Should **uniformly disperse** in compartment
2. Should not leave the compartment
3. Remain **free and unbound** to proteins

Personal Notes

Initial mass of indicator = I (Units in gram as per given)



Injected into the compartment



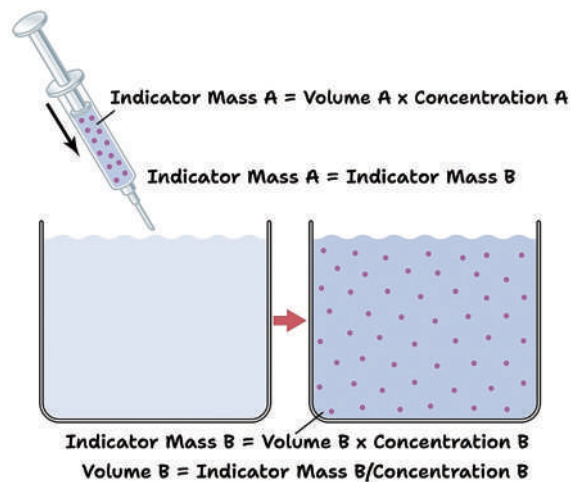
After sometime, sample collected and the concentration of the indicator is measured = C



Thus, assuming the volume of the compartment is much larger than the volume of dye injected

(The usual case of measurement of TBW and ECF)

$$V \text{ (Volume of Compartment)} = I/C$$



- If A (Mass/amount of indicator leaves the compartment. Example: via urinary loss)

$$V = (I-A)/C$$

Indicators Used

Compartment	Indicators
TBW	Deuterium oxide (D_2O), Tritium ($3H_2O$), Aminopyrine
ECF	Non-metabolic saccharides like Inulin (Most common), Mannitol, Sucrose
ICF	Indirectly measured as $ICF = TBW - ECF$
Plasma Volume	Albumin radiolabelled by I-131 isotope
RBC volume	Cr-51 tagging of RBCs.

- Deuterium space = X and mannitol space = Y
- Value of intracellular fluid volume is given as:
 $X - Y$ (X = Deuterium space = TBW, Y = Mannitol space = ECF)
- Cell volume (hematocrit)/packed cell volume: Measured by the Wintrobe and Westergren method)

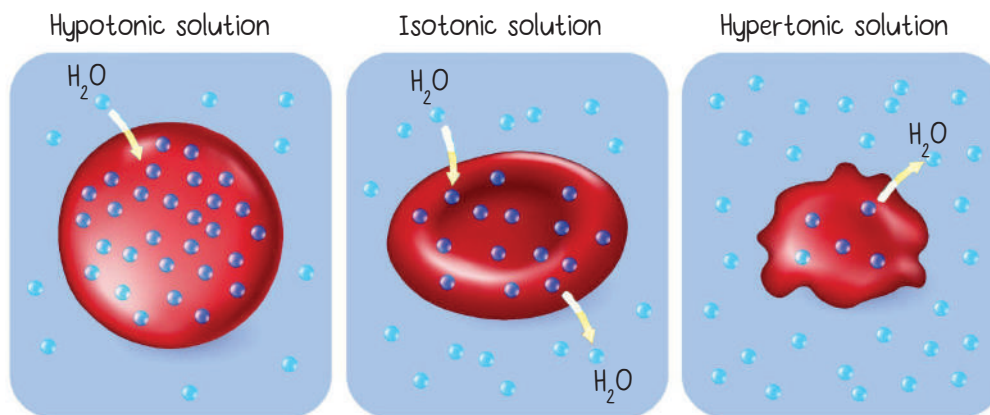
Disturbances in Body Fluid Compartments

- **Dehydration:** Loss of water from the body (Example: Intense exercise - sweating)
- **Over hydration/volume expansion:** Gain of body water (Example: Drinking a lot of water)

Personal Notes

OSMOLARITY AND TONICITY

- It refers to the **osmotic pressure** exerted by a **solution of a solute**, whereas tonicity refers to the capacity of the solution to change volume when in contact with each other compartment of different osmolarity.
- We have two solutions A (Inside RBC) and B (The external solution) in contact with each other, separated by a semipermeable membrane, then the following diagram shows the classification.



Note: The terms, hypertonic, isotonic and hypotonic have no absolute significance but are always significant with respect to other solutions.

- Fluid from the body is primarily lost from ECF (Extracellular Fluid).
- Since, ECF is separated from ICF by a semipermeable cell membrane; **changes in ECF are followed by ICF** i.e., (Changes in intracellular fluid are secondary).
- Normal ECF osmolarity is 286-300 mOsm/L.
- Suppose, a person consumes 10-12 litres of water.
- This water enters the extracellular compartment, but after some time this enters the cells (ICF), causing them to swell by gaining water.

DEHYDRATION

- The classification is with respect to the osmolarity of ECF after fluid loss as compared to the one before fluid loss.

Hypotonic

- Na^+ loss more than water.
- Causes:
 - ▶ Primary hypoaldosteronism
 - ▶ Primary hypoadrenocorticism

Isotonic

- Water and Na^+ lost in equal proportions.
- Causes GI fluid loss like diarrhea, vomiting, burns and hemorrhage.

Hypertonic

- Water lost more than Na^+ .
- Causes:
 - ▶ Diabetes Mellitus (Due to osmotic diuresis by glucose)
 - ▶ Diabetic Insipidus
 - ▶ Chronic Alcoholism

OVERHYDRATION

Classified into 3 Types

- Isotonic
- Hypotonic
- Hypertonic

Isotonic

- Administration of oral/IV saline (0.9%)

Hypotonic

- Drinking water
- Syndrome of Inappropriate ADH secretion (SIADH)

Hypertonic

- Administration of oral/IV hypertonic saline.

- MBBS and MD in Physiology
- Over 29 years of teaching experience
- Author of 3 widely used books on Physiology



Dr. Vivek Nalgirkar

Physiology

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